Paul Leger: Publications (*ORCID:* 0000-0003-0969-5139)

Associate Professor & Software Engineer
Civil Engineering School
Catholic University of the North
Larrondo 1281, Coquimbo, Chile

Email: pleger@gmail.com
Homepage: http://pleger.cl
Pragmatics lab: http://pragmaticslab.com (Director)
LinkedIn: http://cl.linkedin.com/in/plegerm

My research profile IDs are *ORCID* (0000-0003-0969-5139), *Web of Science* (Q-6174-2017), and *Scopus* (36087550600).

NOTE: For conferences/workshops, I use the CORE ranking, which belongs **CO**mputing **RE**search and Education Association of Austral-asia (near to the publication year). CORE is available on http://portal.core.edu.au/conf-ranks.

Journals (Indexed)

Canquil, Diego, Daniel SanMartín, Irene Inoquio-Renteria, and Paul Leger (Jan. 2026). "A Systematic Literature Review on Machine Learning Techniques for Predicting Household Water Consumption". In: *Water Resources Management* 40.2, p. 83. DOI: 10.1007/s11269-026-04496-4. **JCR-WOS**.

Leger, Paul, David Moreno-Lumbreras, Felipe Ruiz, Nicolás Sepúlveda, Ismael Figueroa, and Nicolás Cardozo (Feb. 2026). "The Stackage Repository: An Exploratory Study of its Evolution". In: *Journal of Computer Languages* 86, p. 101386. DOI: 10.1016/j.cola.2026.101386. **JCR-WOS**.

Leger, Paul, Daniel San-Martín, Diego Nuñez, and Tomás Vélez (Feb. 2026). "An Agent-Based Model Reflective Architecture for Social Network Sites". In: *Science of Computer Programming* 252, p. 103452. DOI: 10.1016/j.scico.2026.103452. **JCR-WOS**.

Abdi, Farshid, Shaghayegh Abolmakarem, Amir Karbassi-Yazdi, Paul Leger, Yong Tan, and Giuliani Coluccio (Jan. 2025). "Predicting Patients' Revisit Intention Based on Satisfaction Scores: Combination of Penalized Regression and Neural Networks". In: *IEEE ACCESS* 13, pp. 2783–2800. DOI: 10.1109/ACCESS.2024.3522767. **JCR-WOS**.

Karbassi, Amir, Yang Tan, and Paul Leger (Jan. 2025). "Clustering Financial Institutions in Countries Based on a Hybrid Random Forest and Induced Ordered Weighted Averaging". In: *Journal of Management Analytics*, pp. 1–30. DOI: 10.1080/23270012.2025.2454674. **JCR-WOS**.

Terán, Oswaldo, Paul Leger, and Manuela López (Apr. 2025). "Factors that Drive Market Share and the Oligopolistic Character of Cross-border B2C Ecommerce: An agent-based scenario analysis approach". In: *Simulation: Transactions of the Society for Modeling and Simulation International* 0.0, pp. 1–24. DOI: 10.1177/00375497241296542. **JCR-WOS**.

Junior, Paulo Nocera Alves, Paul Leger, and Isotilia Costa Melo (Oct. 2024). "Efficiency Analysis of Engineering Classes: A DEA Approach Encompassing Active Learning and Expositive Classes Towards Quality Education". In: *Environmental Science & Policy* 160, p. 103856. DOI: 10.1016/j.envsci.2024.103856. **JCR-WOS**.

Leger, Paul, Hiroaki Fukuda, Nicolás Cardozo, and Daniel San Mártin (Jan. 2024). "Exploring a Self-replication Algorithm to Flexibly Match Patterns". In: *IEEE ACCESS* 12, pp. 13553–13570. DOI: 10.1109/ACCESS.2024.3355319. **JCR-WOS**.

B, B. Arinze, DA. Buell, P. Cabezas, JD. Caddell, J. Chase, Y. Chen, E. Chi, SJ. Cosgrove, M. Dabkowski, D. DeVasto, B. Duong, N. Elliot, TS. Ellis, M. Feuer, E. Friess, PB. Gallagher, JF. George, V. Gerard, M. Gertrudix Barrio, G. Getto, G. Giordano, C. Gupta, M. Gupta, V. Gupta, GF. Hayhoe, C. Hidalgo-Alcazar, A. Houser, C. Hubka, KM. Jacobsen, I. Kloo, D. Kong, SJ. Kowalewski, JT. Labriola, A. Lancaster, S. Lang, C. Lauer, J. Lee, Paul Leger, CC. Lewis, C. Liles, and E. Londner (Dec. 2023). "2023 Index IEEE Transactions on Professional Communication Vol. 66". In: *IEEE Transactions on Professional Communication* 66.4, pp. 1–6. DOI: 10.1109/tpc.2023.3338309. **OTHER INDEXES**.

Leger, Paul, Alexandre Bergel, Juan Pablo Sandoval Alcocer, and Leonel Merino (Mar. 2023). "Introduction to Special Issue on Visualization Applied to Software Engineering". In: *Information and Software Technology* 155, p. 107118. DOI: 10.1016/j.infsof.2022.107118. **JCR-WOS. A journal edition**.

Leger, Paul, Nicolás Cardozo, and Hidehiko Masuhara (Apr. 2023a). "An Expressive and Modular Layer Activation Mechanism for Context-Oriented Programming". In: *Information and Software Technology* 156, p. 107132. DOI: 10.1016/j.infsof.2022.107132. **JCR-WOS**.

Leger, Paul, Felipe Ruiz, Nicolás Cardozo, and Hiroaki Fukuda (Feb. 2023). "Benefits, Challenges, and Usability Evaluation of DeloreanJS: A Back-in-Time Debugger for JavaScript". In: *PeerJ Computer Science* 9, e1238. DOI: 10.7717/peerj-cs.1238. **JCR-WOS**.

López, Manuela, Carmen Hidalgo-Alcázar, and Paul Leger (May 2023). "The Effect of Message Repetition on Information Diffusion on Twitter: An Agent-Based Approach". In: *IEEE Transactions on Professional Communication*, pp. 150–169. DOI: 10.1109/TPC.2023.3260449. **JCR-WOS**.

Salinas, Matias, Paul Leger, Hiroaki Fukuda, Nicolás Cardozo, Vannessa Duarte, and Ismael Figueroa (Jan. 2023). "Evaluations of Integrated Programming Environment for First-Year Students in Computer Engineering". In: *Journal of Universal Computer Science* 29.1, pp. 73–98. DOI: 10.3897/jucs.81329. **JCR-WOS**.

Manzano, Carlos, Claudio Meneses, Paul Leger, and Hiroaki Fukuda (Apr. 2022). "An Empirical Evaluation of Supervised Learning Methods for Network Malware Identification Based on Feature Selection". In: *Complexity* 2022.18. DOI: 10.1155/2022/6760920. **JCR-WOS**.

Pizarro, Vicky, Paul Leger, Carmen Hidalgo-Alcázar, and Ismael Figueroa (Jan. 2022). "ABM RoutePlanner: An Agent-Based Model Simulation for Suggesting Preference-Based Routes in Spain". In: *Journal of Simulation* 17.4, pp. 444–461. DOI: 10.1080/17477778.2022.2027826. **JCR-WOS**.

Sosa, Juan Sebastian, Paul Leger, Hiroaki Fukuda, and Nicolás Cardozo (July 2022). "Ad Hoc Systems Management and Specification with Distributed Petri Nets". In: *Journal of Parallel and Distributed Computing* 169, pp. 117–129. DOI: 10.1016/j.jpdc.2022.06.015. **JCR-WOS. NOTE: Selected in "the best of ourselves" on JCC-2022 (main event in Chile)**.

Terán, Oswaldo, Paul Leger, and Manuela López (Mar. 2022). "Modeling and Simulating Chinese Cross-border e-Commerce: An Agent-Based Simulation Approach". In: *Journal of Simulation* 17.6, pp. 1–18. DOI: 10.1080/17477778.2022.2043791. **JCR-WOS**.

Camacho, Marta Cecilia, Francisco Álvarez, César Collazos, Paul Leger, Julián Bermúdez, and Julio Ariel Hurtado (June 2021). "A Collaborative Method for Scoping Software Product Lines: a Case Study in a Small Software Company". In: *Applied Sciences - Basel* 11.15. DOI: 10.3390/app11156820. **JCR-WOS**.

Duarte, Vannessa, Paul Leger, Sergio Contreras, and Hiroaki Fukuda (June 2021). "Using Artificial Neural Network to Detect Fetal Alcohol Spectrum Disorder in Children". In: *Applied Sciences (Basel)* 11.13. DOI: 10.3390/app11135961. **JCR-WOS**.

Figueroa, Ismael, Paul Leger, and Hiroaki Fukuda (Jan. 2021). "Which Monads Haskell Developers Use: An Exploratory Study". In: *Science of Computer Programming* 201, p. 102523. DOI: 10.1016/j.scico.2020.102523. **JCR-WOS**.

Fukuda, Hiroaki, Ryota Gunji, Tadahiro Hasegawa, Paul Leger, and Ismael Figueroa (Feb. 2021). "DSSM: Distributed Streaming data Sharing Manager". In: *Sensors* 21.4, pp. 1–15. DOI: 10.3390/s21041344. **JCR-WOS**.

Leger, Paul, Hiroaki Fukuda, and Ismael Figueroa (Sept. 2021). "Continuations and Aspects to Tame Callback Hell on the Web". In: *Journal of Universal Computer Science* 27.9, pp. 955–978. DOI: 10.3897/jucs.72205. **JCR-WOS**.

Aguad, Yaria, Natalia Araya, Javiera Elías, Carmen Hidalgo-Alcázar, and Paul Leger (Dec. 2020). "Impacto del Etiquetado Nutricional en el Comportamiento de los Consumidores de la IV Región, Chile". In: *Revista de Investigación Aplicada en Ciencias Empresariales (RIACE)* 9.1, pp. 7–26. DOI: 10.22370/riace.2020.9.1.2596. **LATINDEX**.

Gálvez, Mará, Paullette Salinas, Camilo San-Martín, Paul Leger, and Carmen Hidalgo-Alcázar (Dec. 2020). "Caso de Estudio Sobre la Efectividad en la Utilización de Facebook en la Venta de Productos y Servicios en Empresas de la Región de Coquimbo". In: *Revista de Investigación Aplicada en Ciencias Empresariales (RIACE)* 9.1, pp. 27–48. DOI: 10.22370/riace.2020.9.1.2597. **LATINDEX**.

Araya, Natalia, Paul Leger, and Manuela López (Apr. 2019). "Whom do I Choose to Diffuse Information on Twitter? An Agent-Based Model Approach". In: *IEEE Latin America Transactions* 17.4, pp. 677–683. DOI: 10.1109/TLA.2019.8891933. **JCR-WOS**.

Figueroa, Ismael, Cristhy Jiménez, Hector Allende-Cid, and Paul Leger (July 2019). "Developing Usability Heuristics with PROMETHEUS: A Case Study in Virtual Learning Environments". In: *Computer Standards & Interfaces* 65, pp. 132–142. DOI: 10.1016/j.csi.2019.03.003. **JCR-WOS**.

Rojas, Felipe, Andrea Fernández, Paulina Gutierrez, and Paul Leger (Dec. 2019). "Identificando Brechas de Uso de Sistemas de Información (SIA) para Micro, Pequeñas y Medianas Empresas en La Serena y Coquimbo". In: *Revista de Investigación Aplicada en Ciencias Empresariales (RIACE)* 8.1, pp. 77–113. DOI: 10.22370/riace.2019.8.1.2073. **LATINDEX**.

Fukuda, Hiroaki, Paul Leger, and Namiki Keita (Aug. 2016). "CMSN: An Efficient and Effective Agent Lookup for Mobile Agent Middleware". In: *Journal of Universal Computer Science* 22.8, pp. 1072–1096. DOI: 10.3217/jucs-022-08-1072. **JCR-WOS**.

Leger, Paul, Manuela López, Carmen Hidalgo-Alcázar, and Hiroaki Fukuda (May 2016). "An Open Agent-Based Model to Simulate the Effect of WOM Marketing Campaigns". In: *EAI Endorsed Transactions on Smart Cities* 1.4. ISSN: 2518-3893. DOI: 10.4108/eai.3-12-2015.2262533.

Rocha, Wiliam, Hiroaki Fukuda, and Paul Leger (May 2016). "Modular Asynchronous Web Programming: Advantages and Challenges". In: *EAI Endorsed Transactions on Collaborative Computing* 2.8. ISSN: 2312-8623. DOI: 10.4108/eai.3-12-2015.2262472.

Vidal, Cristian, David Benavides, Paul Leger, José Galindo, and Hiroki Fukuda (May 2016). "Mixing of Join Point Interfaces and Feature-Oriented Programming for Modular Software Product Line". In: *EAI Endorsed Transactions on Scalable Information Systems* 3.10. DOI: 10.4108/eai.3-12-2015.2262534. **JCR-WOS**.

Fukuda, Hiroaki and Paul Leger (June 2015d). "SyncAS: A Virtual Block Approach to Tame Asynchronous Programming". In: *International Journal of Software Engineering and Knowledge Engineering (IJSEKE)* 25.5, pp. 887–907. DOI: 10.1142/S0218194015400252. **JCR-WOS**.

Leger, Paul, Éric Tanter, and Hiroaki Fukuda (May 2015). "An Expressive Stateful Aspect Language". In: *Science of Computer Programming* 102.0, pp. 108–141. DOI: 10.1016/j.scico.2015.02.001. **JCR-WOS**.

Liu, Shih-Hsi, Junichi Suzuki, Paul Leger, Hiroaki Fukuda, and Marjan Mernik (June 2015). "Featured Topic Issue - Emerging Synergies of Artificial Intelligence and Software Engineering". In: *International Journal of Software Engineering and Knowledge Engineering (IJSEKE)* 25.5, pp. 803–805. DOI: 10.1142/S0218194015020040. **JCR-WOS. A journal edition**.

Leger, Paul, Grecia Gálvez, Lino Cubillos, Diego Cosmelli, Milton Inostroza, Éric Tanter, Gina Luci, and Jorge Soto Andrade (Mar. 2014). "ECOCAM, un sistema computacional adaptable al contexto para promover estrategias de cálculo mental: características de su diseño y resultados preliminares". In: *Revista Latinoamericana de Investigación en Matemática Educativa* 17.1, pp. 33–58. DOI: 10.12802/relime.13.1712. **JCR-WOS**.

Ramírez, Pablo, Paul Leger, and Andres Vallone (July 2014). "Un Modelo Flexible para la Simulación de Distribución de Ciudades". In: *Revista chilena de ingeniería* 22.3, pp. 363–373. DOI: 10.4067/S0718-33052014000300007. **SCOPUS**.

Ramírez, Patricio, Paul Leger, and Carolina Fuentes (Nov. 2014). "Los Programas de Ingeniería en Sistemas de Información y Control de Gestión en Chile: Un Análisis Cuantitativo". In: *Formación Universitaria* 7.2, pp. 21–28. DOI: 10.4067/S0718-50062014000200004. **SCOPUS**.

Leger, Paul, Éric Tanter, and Rémi Douence (Sept. 2013). "Modular and Flexible Causality Control on the Web". In: *Science of Computer Programming* 78.9, pp. 1538–1558. DOI: 10.1016/j.scico.2012.11.005. **JCR-WOS**.

Gálvez, Grecia, Diego Cosmelli, Lino Cubillos, Paul Leger, Arturo Mena, Éric Tanter, Ximena Flores, Gina Luci, Soledad Montoya, and Jorge Soto Andrade (Mar. 2011). "Estrategias Cognitivas para el Cálculo Mental". In: *Revista Latinoamericana de Investigación en Matemática Educativa* 14.1, pp. 9–40. **JCR-WOS**.

Conferences (SCOPUS)

Liu, Shijin, Hiroaki Fukuda, and Paul Leger (Mar. 2026). "SOM-based Continuous XGBoost Approach for DDoS Attack Detection and Mitigation in Edge SDN". In: *Workshop on Programmable and Learning Architectures for Intelligent Networks*. Athens, Greece. **SCOPUS. Rank National-France (CORE ranking). To appear**.

Ogino, Toshihiro, Hiroaki Fukuda, and Paul Leger (July 2025). "Stepwise Distributed Execution for Load Reduction in Machine Learning Systems". In: *International Symposium on Parallel and Distributed Computing (ISPDC)*. Rennes, France. **To appear. SCOPUS. Rank C (CORE ranking)**.

You, Koshi, Hiroaki Fukuda, and Paul Leger (Apr. 2025). "The Impact of Context-Oriented Programming on Declarative UI Design in React". In: *Evaluation of Novel Approaches to Software Engineering (ENASE)*. Porto, Portugal, pp. 700–707. DOI: 10.5220/0013432300003928. **SCOPUS. Rank B (CORE ranking)**.

Nuñez, Diego, Tomás Vélez, Paul Leger, and Daniel San Martín (Apr. 2024). "A Reflective Architecture for Agent-Based Models Applied to Social Network Sites". In: *International Conference on Enterprise Information Systems (ICEIS)*. Angers, France: IEEE, pp. 981–988. DOI: 10.5220/0012615100003690. **SCOPUS**.

Cardenas, Sergio, Paul Leger, Hiroaki Fukuda, and Nicolás Cardozo (July 2023). "Points-to Analysis for Context-Oriented JavaScript Programs". In: *Proceedings of the 25th ACM International Workshop on Formal Techniques for Java-like Programs (FTfJP 23)*. Seattle, USA, pp. 18–24. DOI: 10.1145/3605156.3606451. **SCOPUS. Rank C (CORE ranking)**.

Leger, Paul, Nicolás Cardozo, and Hidehiko Masuhara (Oct. 2023b). "An Expressive and Modular Layer Activation Mechanism for Context-Oriented Programming". In: *39th IEEE International Conference on Software Maintenance and Evolution – Journal First Track*. Bogota, Colombia: IEEE. **Rank A (CORE ranking). NOTE: This paper was published first in the Journal Information & Technology (IST); now it is accepted as Journal First**.

Liu, Shijin, Hiroaki Fukuda, and Paul Leger (Nov. 2023a). "A RF-based Low Rate DDoS Attack Real-time Detection System". In: *33rd International Telecommunication Networks and Applications Conference (ITNAC)*. Melbourne, Australia: IEEE, pp. 304–309. DOI: 10.1109/ITNAC59571.2023.10368543. **SCOPUS. Rank C-Australasian (CORE ranking)**.

— (Mar. 2023b). "Real-time DDoS Attack Defense System in SDN Using LSSOM". In: *26th Conference on Innovation in Clouds, Internet and Networks and Workshops (ICIN)*. Paris, France, pp. 69–73. DOI: 10.1109/ICIN56760.2023.10073509. **SCOPUS. Rank National-France (CORE ranking)**.

Leger, Paul, Felipe Ruiz, Nicolás Sepúlveda, and Ismael Figueroa (Apr. 2022). "Evolution of a Haskell Repository and its Use of Monads: An Exploratory Study of Stackage". In: *Proceedings of the 37th Annual ACM Symposium on Applied Computing (SAC 2022)*. Virtual Conference: ACM Press, pp. 1475–1482. DOI: 10.1145/3477314.3506982. **SCOPUS. Rank B (CORE ranking)**.

Fukuda, Hiroaki, Ryota Gunji, Tadahiro Hasegawa, Paul Leger, and Ismael Figueroa (Jan. 2020). "Toward distributed Streaming data Sharing Manager for Autonomous Robot Control". In: *IEEE/SICE International Symposium on System Integration (SII)*. Honolulu, USA: IEEE, pp. 862–866. DOI: 10.1109/SII46433.2020.9025920. **SCOPUS. NOTE: Superseded by Sensors**.

Fukuda, Hiroaki, Paul Leger, and Ismael Figueroa (May 2020). "A Practical Methodology to Learn Computer Architecture, Assembly Language, and Operating System". In: *Proceedings of the 12th International Conference on Computer Supported Education (CSEDU)*. Prague, Czech Republic: ACM Press, pp. 333–340. DOI: 10.5220/0009319503330340. **SCOPUS**.

Jaque, Dayana, Paul Leger, Felipe Machorro, and Vanessa Romero Ortiz (Nov. 2020). "Indicadores de Capital Intelectual para Gestionar el Desempeño en Universidades Chilenas". In: *36th IBIMA International Conference*. Granada, España: IBIMA.

Leger, Paul, David Contreras, Orietta Nicolis, Carlos Hernández, Patricio Rojas, Giovanni Giachetti, Eduardo Quiroga, Loreto Telgie, Oriel Herrera, Lioubov Dombrovskaia, and Alonso Inostrosa-Psijas (Nov. 2020). "SCCC 2020 Opinion". In: *2020 39th International Conference of the Chilean Computer Science Society (SCCC)*. Coquimbo, Chile: IEEE, pp. 1–1. DOI: 10.1109/SCCC51225.2020.9281281. **SCOPUS. NOTE: Conference Proceedings**.

Machorro, Felipe, Vanessa Romero Ortiz, Paul Leger, and Dayana Jaque (Nov. 2020). "Efecto Mediador de la Gestión del Conocimiento entre las Tecnologías de Información y el Desempeño Organizacional de las Instituciones de Educación Superior". In: *36th IBIMA International Conference*. Granada, España: IBIMA.

Manzano, Carlos, Claudio Meneses, and Paul Leger (Nov. 2020). "Empirical Comparison of Supervised Algorithms for Ransomware Identification on Network Traffic". In: *2020 - The International Conference of the Chilean Computer Science Society (SCCC)*. Coquimbo, Chile: IEEE, pp. 1–7. DOI: 10.1109/SCCC51225.2020.9281283. **SCOPUS**.

Pizarro, Vicky, Mauricio Gallardo, and Paul Leger (Nov. 2020). "Identificando Distribuciones Condicionales Del Ingreso Personal a Través de Técnicas de Agrupamiento". In: *36th IBIMA International Conference*. Granada, España: IBIMA.

Figueroa, Ismael, Bruno García, and Paul Leger (Sept. 2018). "Towards Progressive Verification in Dafny". In: *Proceedings of the XXII Brazilian Symposium on Programming Languages (SBLP 2018)*. São Carlos, Brazil: ACM Press, pp. 90–97. DOI: 10.1145/3264637.3264649. **SCOPUS**.

Leger, Paul and Hiroaki Fukuda (Apr. 2017). "Sync/CC: Continuations and Aspects to Tame Callback Dependencies on JavaScript Handlers". In: *Proceedings of the 32nd Annual ACM Symposium on Applied Computing (SAC 2017)*. Marrakech, Morocco: ACM Press, pp. 1245–1250. DOI: 10.1145/3019612.3019783. **Rank B (CORE ranking). NOTE: Superseded by JUCS (Journal of Universal Computer Science)**.

Sabin, Mihaela, Barbara Viola, John Impagliazzo, Renzo Angles, Mariela Curiel, Paul Leger, Jorge Murillo, Hernán Nina, José Antonio Pow-Sang, and Ignacio Trejos (July 2016). "Latin American Perspectives to Internationalize Undergraduate Information Technology Education". In: *Proceedings of the 2016 ACM Conference on Innovation and Technology in Computer Science Education (ITiCSE 2016)*. Arequipa, Peru: ACM Press, pp. 1–22. DOI: 10.1145/3024906.3029847. **SCOPUS. Rank A (CORE ranking). NOTE: Selected in "the best of ourselves" Chilean JCC-2019**.

Rocha, Wiliam, Hiroaki Fukuda, and Paul Leger (Dec. 2015). "Modular Asynchronous Web Programming: Advantages and Challenges". In: *Proceedings of 9th International Conference on Bio-inspired Information and Communications Technologies (BICT 2015)*. ACM Press. DOI: 10.4108/eai.3-12-2015.2262472. **SCOPUS**.

Vidal, Cristian, David Benavides, Paul Leger, José Galindo, and Hiroki Fukuda (Dec. 2015). "Mixing of Join Point Interfaces and Feature-Oriented Programming for Modular Software Product Line". In: *Proceedings of 9th International Conference on Bio-inspired Information and Communications Technologies (BICT 2015)*. New York, USA: ACM Press. DOI: 10.4108/eai.3-12-2015.2262534. **SCOPUS. NOTE: A mirror version (with the same DOI) was published in the journal EAI Endorsed Transactions on Scalable Information Systems**.

Leger, Paul and Éric Tanter (Sept. 2010). "An Open Trace-Based Mechanism". In: *Proceedings of the 14th Brazilian Symposium on Programming Languages (SBLP 2010)*. Ed. by Jonathan Aldrich and Ricardo Massa. Salvador - Bahia, Brazil, pp. 123–138.

Toledo, Rodolfo, Paul Leger, and Éric Tanter (Mar. 2010). "AspectScript: Expressive Aspects for the Web". In: *Proceedings of the 9th ACM International Conference on Aspect-Oriented Software Development (AOSD 2010)*. Rennes and Saint Malo, France: ACM Press, pp. 13–24. DOI: 10.1145/1739230.1739233. **SCOPUS. Rank A (CORE ranking)**.

Workshops (SCOPUS)

Fukuda, Hiroaki, Paul Leger, and Nicolás Cardozo (June 2022). "Layer Activation Mechanism for Asynchronous Executions in JavaScript". In: *Proceedings of the 14th International Workshop on Context-Oriented Programming and Advanced Modularity (COP 22)*. Berlin, Germany: ACM Press, pp. 1–8. DOI: 10.1145/3570353.3570354. **SCOPUS**.

Cleveland, Margareth and Paul Leger (Nov. 2020). "A Collaborative Learning Strategy in an MIS Development Course Using Case Method in Engineering in Information and Management Control". In: *Congreso Chileno de TICs para la Educación (TICXED)*. Coquimbo, Chile: IEEE, pp. 1–5. DOI: 10.1109/SCCC51225.2020.9281192. **SCOPUS**.

Leger, Paul, Hidehiko Masuhara, and Ismael Figueroa (July 2020). "Interfaces for Modular Reasoning in Context-Oriented Programming". In: *Proceedings of the 12th International Workshop on Context-Oriented Programming and Advanced Modularity (COP 20)*. 3. Virtual Event, USA: ACM Press, pp. 1–7. DOI: 10.1145/3422584.3423152. **SCOPUS**.

Leger, Paul and Hiroaki Fukuda (Mar. 2016). "Using Continuations and Aspects to Tame Asynchronous Programming on the Web". In: *Proceedings of the Workshop on Foundations of Aspect-Oriented Languages (FOAL 2016)*. Malaga, Spain: ACM Press, pp. 79–82. DOI: 10.1145/2892664.2892675. **SCOPUS. Rank C (CORE ranking). NOTE: Superseded by JUCS (Journal of Universal Computer Science)**.

Fukuda, Hiroaki and Paul Leger (June 2015c). "Proposals for Modular Asynchronous Web Programming: Issues and Challenges". In: *Proceedings on Pervasive Web Technologies, Trends and Challenges (PEWET 2015)*. Vol. 9396. Lecture Notes in Computer Science. Rotterdam, the Netherlands: Springer, pp. 91–

102. DOI: 10.1007/978-3-319-24800-4_8. **SCOPUS. NOTE: Published on Lectures Notes in Computer Science**.

Silva, Cristian Vidal, David Benavides, José Ángel Galindo, Paul Leger, Rodolfo Villarroel, and Sebastián Valenzuela (Nov. 2015). "JPI Feature Models - Exploring a JPI and FOP Symbiosis". In: *WASE 2015 Workshop on Advanced Software Engineering*. Santiago, Chile: IEEE, pp. 1–6. DOI: 10.1109/SCCC.2015.7416583.

Leger, Paul and Éric Tanter (Mar. 2012). "A Self-Replication Algorithm to Flexibly Match Execution Traces". In: *Proceedings of the 11th Workshop on Foundations of Aspect-Oriented Languages (FOAL 2012)*. Potsdam, Germany: ACM Press, pp. 27–32. DOI: 10.1145/2162010.2162019. **SCOPUS. Rank C (CORE ranking)**.

Congresses (*Conferences/Workshops not indexed*)

Note: These papers are just presentations of technical reports or working papers.

Moreno-Lumbreras, David, Paul Leger, Sergio Montes-León, Jesus Gonzalez-Barahona, and Gregorio Robles (Sept. 2025). "Evaluating the Comprehension of the Stackage Ecosystem: A Comparison between VR and 2D Visualizations". In: *Registered Report: 2025 IEEE International Conference on Software Maintenance and Evolution (ICSME)*. Auckland, New Zealand. DOI: 10.2139/ssrn.5362665. **Registered Report**.

Espinoza, Francis and Paul Leger (Dec. 2024). "Using Artificial Intelligence to Democratise the Information". In: *International Conference Promoting Democracy and the Rule of Law – Global Perspectives Santiago de Chile*. Santiago, Chile.

Ortiz, Mario, Paul Leger, and Carmen Hidalgo-Alcázar (Sept. 2022). "Clasificación de Usuarios en Perfiles de Compradores en el Sector Inmobiliario: Análisis Mediante una Experiencia Virtual". In: *XXXIII Congreso de Marketing (AEMARK 2022)*. Valencia, Spain.

López, Manuela, Oswaldo Terán, and Paul Leger (Sept. 2021). "Modeling Chinese Cross-Border B2C E-commerce in Spain: A Multi-Agent Based Simulation Approach". In: *XXXII Congreso de Marketing (AEMARK 2021)*. Baeza y Úbeda, Spain.

Figueroa, Ismael, Paulina Guitierrez, Natalia Araya, and Paul Leger (Nov. 2020). "Estudio Indagatorio sobre la Usabilidad Percibida de los Sistemas de Información Contable y Financieros". In: *Conferencia Académica Permanente de Investigación Contable (CAPIC)*. La Serena, Chile.

Pinto, Domingo, Diego Cortes, Paul Leger, and Manuela López (Sept. 2020). "Una Arquitectura de una Implementación Abierta de un Modelo Basado en Agentes para Campañas de WOM Marketing". In: *XI Congreso Internacional de Computación e Informática del Norte de Chile (Infonor 2020)*. Atacama, Chile.

Leger, Paul, Felipe Ruiz, and Guillermo Victorero (Aug. 2019). "DeloreanJs: Un Debugger en el Tiempo para JavaScript". In: *X Congreso Internacional de Computación e Informática del Norte de Chile (Infonor 2019)*. Antofagasta, Chile.

López, Manuela, Carmen Hidalgo-Alcázar, and Paul Leger (Sept. 2019). "Don't Wear me Out! The Effect of Tweet Repetition". In: *XXXI Congreso de Marketing (AEMARK 2019)*. Cáceres, Spain. **Superseded by IEEE Transactions on Professional Communication**.

Pizarro, Vicky, Paul Leger, Carmen Hidalgo-Alcázar, and Ismael Figueroa (Sept. 2018). "Simulación de un Modelo Basado en Agentes para la Elección de Destinos Turísticos". In: *IX Congreso Internacional de Computación e Informática del Norte de Chile (Infonor 2018)*. Iquique, Chile.

Araya, Natalia, Paul Leger, and Manuela López (Sept. 2017). "A Quién Selecciono? Análisis De Hubs y Líderes de Opinión en la Difusión de Información en Twitter. Una Simulación Basada en Agentes". In: *XXIX Congreso Internacional de Marketing (AEMARK 2017)*. Sevilla, Spain. **Superseded by IEEE Latin America Transactions**.

López, Manuela, Carmen Hidalgo-Alcázar, and Paul Leger (Sept. 2017a). "How many times should a company repeat a tweet to reach the highest diffusion? An agent-based simulation approach". In: *V Workshop De Jóvenes Investigadores En Economía Y Empresa*. Jaca, Spain.

— (July 2017b). "The Effect of Message Repetition on Information Diffusion on Twitter Using an Agent-Based Simulation". In: *The 2017 International Conference on Research in Advertising (ICORIA 2017)*. Ghent, Belgium. **Superseded by International Journal of Advertising**.

Pizarro, Vicky, Paul Leger, and Carmen Hidalgo-Alcázar (Sept. 2017). "Elección de los Destinos Turísticos de España: Una Simulación Basada en Agentes". In: *XXIX Congreso Internacional de Marketing (AEMARK 2017)*. Sevilla, Spain.

López, Manuela, Carmen Hidalgo-Alcázar, and Paul Leger (Sept. 2016). "Tell Me Again! The Effect of Message Repetition on Information Diffusion on Twitter: An Agent-Based Approach". In: *XXVIII Congreso de Marketing (AEMARK 2016)*. Leon, Spain.

— (Sept. 2015a). "An Agent-based Information Diffusion Study on Social Network Sites". In: *III Workshop Jóvenes Investigadores en Economía y Empresa*. Teruel, Spain. **Superseded by International Journal of Advertising**.

— (Sept. 2015b). "Don't Bother Me with Many Messages: An Information Diffusion Study on Social Network Sites with An Agent-Based Approach". In: *XXVII Congreso de Marketing (AEMARK 2015)*. Pamplona, Spain. **Best Working Paper Award!. Superseded by International Journal of Advertising**.

Vidal, Cristian, David Benavides, José Galindo, and Paul Leger (Sept. 2015). "Exploring the Synergies between Join Point Interfaces and Feature-Oriented Programming". In: *XX Jornadas de Ingeniería del Software y Bases de Datos (JISBD 2015)*. Santander, España.

Azegami, Takeshi, Hiroaki Fukuda, and Paul Leger (Nov. 2014). "Towards a Virtual Block Approach to Tame Asynchronous Programming". In: *Proceedings of 8th International Conference on Bio-inspired Information and Communications Technologies (BICT 2014)*. Boston, USA, pp. 239–242. DOI: 10.4108/icst.bict.2014.257939. **Superseded by IJSEKE (International Journal of Software Engineering and Knowledge Engineering)**.

Vidal, Elizabeth, Edwin Morales, and Paul Leger (Oct. 2014). "Usando BPMN para Modelar Procesos en el Área de Ingeniería y Proyectos de una Empresa Minera del Perú". In: *VI Congreso Internacional de Computación y Telecomunicaciones (COMTEL 2014)*. Lima, Peru.

Ramírez, Pablo, Paul Leger, and Andres Vallone (Aug. 2013). "Un Modelo Flexible para la Simulación de Distribución de Ciudades". In: *IV Congreso Internacional de Computación e Informática del Norte de Chile (Infonor 2013)*. Coquimbo, Chile. **Superseded by INGENIARE**.

Ramírez, Patricio, Paul Leger, and Carolina Fuentes (Nov. 2013). "Explorando los Programas de Ingeniería en Sistemas de Información y Control de Gestión en Chile". In: *XV Congreso Chileno de Educación Superior en Computación (CCESC 2013)*. Temuco, Chile. **Superseded by Formación Universitaria**.

Leger, Paul (Nov. 2010). "Developing Context-Aware Systems Using Expressive Trace-based Mechanisms". In: *Encuentro Tesistas de Doctorados en Jornada Chilena de Computación (JCC)*. Antofagasta, Chile.

Leger, Paul and Éric Tanter (Mar. 2010). "Towards an Open Trace-Based Mechanism". In: *Proceedings of the Ninth Workshop on Foundations of Aspect-Oriented Languages (FOAL 2010)*. Ed. by Gary T. Leavens, Shmuel Katz, and Mira Mezini. Rennes and Saint-Malo, France, pp. 25–30. **Tech report CS-TR-10-04, University of Central Florida**.

Leger, Paul (Nov. 2008). "Context-Dependent Adaptations Using Tracematch Extensions". In: *Encuentro Tesistas de Doctorados en Jornada Chilena de Computación (JCC)*. Punta Arenas, Chile. **Published in http://lahuen.dcc.uchile.cl/ jcc2008/es/trabajos.html**.

Ph. D.

Leger, Paul (Nov. 2012). "Opening Up Trace-Based Mechanisms — Application to Context-Aware Systems". PhD thesis. University of Chile, pp. 1–180. **Published in the University of Chile's system**.